European Genomic Data Infrastructure

European Genomic Data Infrastructure (GDI)

ELIXIR All Hands 2023, 5-8 June, Dublin, Ireland

Advancing Data-driven Biomedical Research and Personalised Medicine Solutions to Benefit Citizens of Europe

The Genomic Data Infrastructure (GDI) project - co-funded by the Digital Europe Programme and coordinated by ELIXIR - brings together national agencies, research organisations, and technology providers in 20 countries to provide a cross-border federated network of national genome collections, associated with other relevant data.

GDI

Supports the 1+Million Genomes (1+MG) initiative ambition to enable secure access to high-quality genomics and the corresponding clinical data across Europe for better research, personalised healthcare and health policy making,

Establishes a **federated, sustainable and secure infrastructure based on open community standards** to access genomic and related phenotypic and clinical data across Europe,

Builds on the Beyond 1 Million Genomes (B1MG) project outputs, including technical specifications and implementation guidelines that capture European best practice.

1+MG Initiative

"1+MG facilitates countries to realise a practice of personalised medicine and health, based upon a shared 'trust framework' and the infrastructure to safely access and integrate high quality genomic data and other health data across borders."

Ambition and Achievements of the Project Pillars

Pillar I: Long-term sustainability

Ensure the infrastructure continues to operate after the end of the GDI project and make data accessible across Europe

Achievements
November 2022-June 2023

- 18 countries have submitted an Expression of Interest to constitute the European Digital Infrastructure Consortium.

Pillar II: Infrastructure deployment

Realise the 1+MG infrastructure: ensure European data resources can operate as part of the 1+MG infrastructure

Achievements
November 2022-June 2023

- Starter Kit:
  - Reference implementations are available for deployment;
  - GDI Nodes have validated infrastructure, using 3,000 realistic synthetic genomes and associated phenotypic data;
- Nodes have mapped to Technology Readiness Levels;
- Draft data management policy with ELSI best practices has been produced.

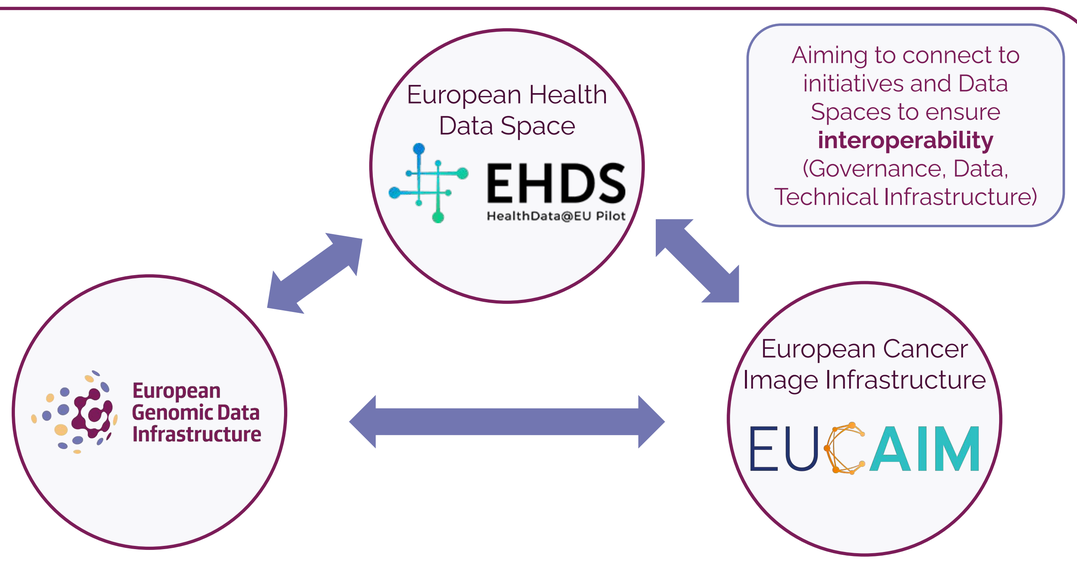
Pillar III: Use cases

Provide innovative solutions for clinicians, researchers and innovators

Achievements
November 2022-June 2023

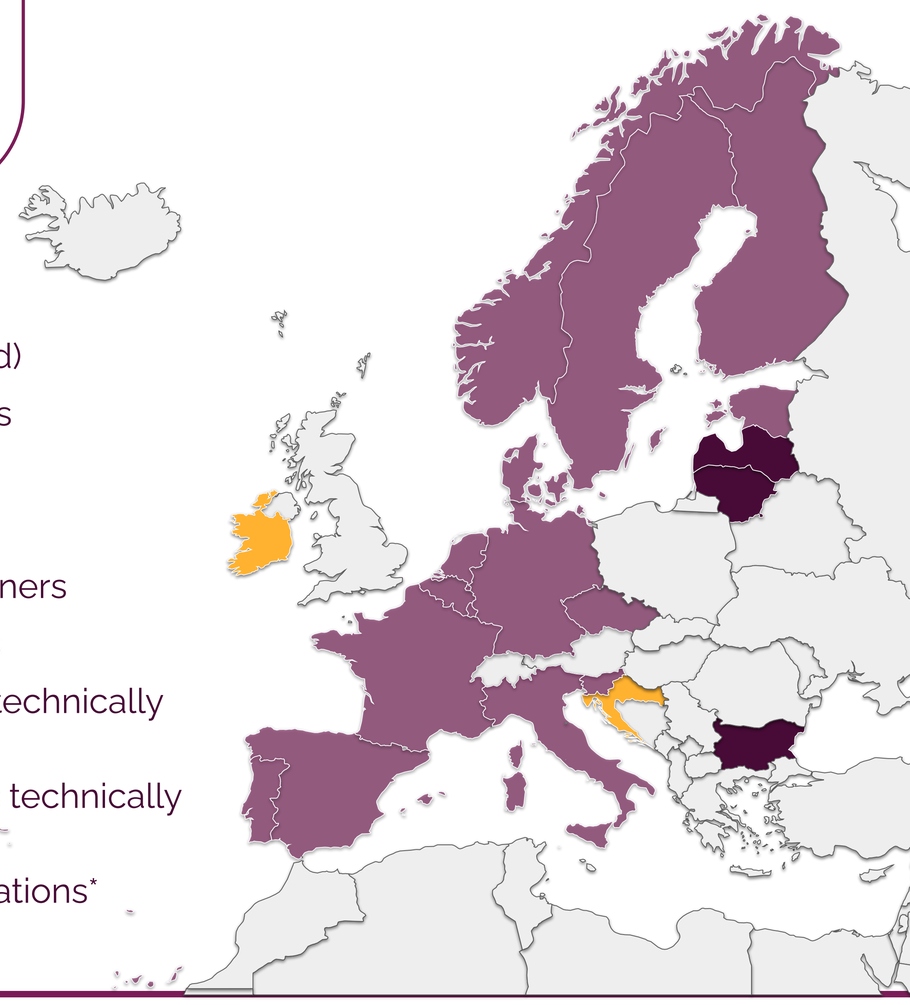
- The initial process to validate distributed learning models has been defined.

Connection to relevant EU Initiatives



GDI in numbers

€40M (50% co-funded)
45 Project Beneficiaries
6 Affiliated Entities
7 Associated Partners
TOTAL: **58** Project Partners
20 European countries
At least **6** countries technically operational by 2024
At least **15** countries technically operational by 2026
2 International Organisations*
*EMBL and BBMRI-ERIC



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